GRADE 9 INTEGRATED SCIENCE
FULL-YEAR SCHEME OF WORK

Academic Year: _____ **School:** _____ **Teacher:** _____

Planning note: This is an original teacher-prepared scheme aligned to the Grade 9 KICD Integrated Science curriculum design. The curriculum design suggests 150 lessons. The school should adapt weekly pacing to its approved calendar, learner progress, available laboratory facilities and safety requirements.

TERM 1 — 55 planned curriculum lessons

Suggested school-calendar window: 14 weeks. Use remaining timetable periods for revision, CATs, practical work, projects, remediation, enrichment, examinations, feedback and school-based activities.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
1	1	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
1	2	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the	The learner is guided to	How does the arrangement of subatomic particles	KICD curriculum design; learner's book; teacher-prepared	Oral questions; observation; practical skills checklist;	

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				learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	help us describe an atom?	notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
1	3	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate,	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
1	4	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				scientific ideas and handling learning models.					
1	5	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
2	1	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
2	2	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				learner should show curiosity and accuracy when representing scientific ideas and handling learning models.					
2	3	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
2	4	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
2	5	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	supported by evidence.				
3	1	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
3	2	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to:	The learner is guided to observe, explore and investigate	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written	

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				1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		specimens; laboratory apparatus and safety equipment where appropriate.	exercises; group tasks; peer/self-assessment; exit ticket.	
3	3	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling learning models.	appropriate; discuss findings with peers; and present explanations supported by evidence.				
3	4	Mixtures, Elements and Compounds	Structure of the Atom	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe the basic structure of the atom and relate subatomic particles to atomic identity and simple representations. 2. Skills: The learner should be able to investigate, represent and explain ideas related to structure of the atom using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show curiosity and accuracy when representing scientific ideas and handling	The learner is guided to observe, explore and investigate structure of the atom using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the arrangement of subatomic particles help us describe an atom?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				learning models.					
3	5	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
4	1	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	appropriate; discuss findings with peers; and present explanations supported by evidence.				
4	2	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
4	3	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
4	4	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				and care for materials and the environment.					
4	5	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
5	1	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
5	2	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
5	3	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to	The learner is guided to observe, explore and investigate metals and alloys using suitable models,	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
5	4	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

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				handling, responsible use and care for materials and the environment.					
5	5	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
6	1	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
6	2	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
6	3	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge:	The learner is guided to observe, explore and investigate metals and alloys	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		apparatus and safety equipment where appropriate.	peer/self-assessment; exit ticket.	
6	4	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				learner should demonstrate safe handling, responsible use and care for materials and the environment.	evidence.				
6	5	Mixtures, Elements and Compounds	Metals and Alloys	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate selected physical properties and uses of metals and alloys. 2. Skills: The learner should be able to investigate, represent and explain ideas related to metals and alloys using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate safe handling, responsible use and care for materials and the environment.	The learner is guided to observe, explore and investigate metals and alloys using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	Why are different metals and alloys selected for particular purposes?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
7	1	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
7	2	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				and use resources responsibly.					
7	3	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
7	4	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water.	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
7	5	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				resources responsibly.					
8	1	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
8	2	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
8	3	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				responsibly.					
8	4	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
8	5	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
9	1	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
9	2	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
9	3	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
9	4	Mixtures, Elements and Compounds	Water Hardness	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain temporary and permanent hardness of water and investigate methods used to soften hard water. 2. Skills: The learner should be able to investigate, represent and explain ideas related to water hardness using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate safe water practices and use resources responsibly.	The learner is guided to observe, explore and investigate water hardness using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How can we identify hard water and make it more suitable for use?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
9	5	Living Things and	Nutrition in Plants	At the end of the	The learner is	What conditions do	KICD curriculum	Oral questions;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
		Environment		lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	green plants require to manufacture food?	design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
10	1	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations,	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	present explanations supported by evidence.				
10	2	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
10	3	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
10	4	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
10	5	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to:	The learner is guided to observe, explore and investigate	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		specimens; laboratory apparatus and safety equipment where appropriate.	exercises; group tasks; peer/self-assessment; exit ticket.	
11	1	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				learner should appreciate the importance of plants and care for the environment.	evidence.				
11	2	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
11	3	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
11	4	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
11	5	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to	The learner is guided to observe, explore and investigate nutrition in plants using suitable models,	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				

Teacher's reflection on the term: _____

_____

TERM 2 — 55 planned curriculum lessons

Suggested school-calendar window: 13 weeks. Use remaining timetable periods for revision, CATs, practical work, projects, remediation, enrichment, examinations, feedback and school-based activities.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
1	1	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
1	2	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
1	3	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
1	4	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				environment.					
1	5	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
2	1	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate;	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	discuss findings with peers; and present explanations supported by evidence.				
2	2	Living Things and Environment	Nutrition in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain processes and requirements involved in plant nutrition. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should appreciate the importance of plants and care for the environment.	The learner is guided to observe, explore and investigate nutrition in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	What conditions do green plants require to manufacture food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
2	3	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
2	4	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				things and appreciate biological diversity.					
2	5	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
3	1	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
3	2	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
3	3	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
3	4	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	where appropriate; discuss findings with peers; and present explanations supported by evidence.				
3	5	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
4	1	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to:	The learner is guided to observe, explore and investigate	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		specimens; laboratory apparatus and safety equipment where appropriate.	exercises; group tasks; peer/self-assessment; exit ticket.	
4	2	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	present explanations supported by evidence.				
4	3	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
4	4	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens,	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
4	5	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence.	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.					
5	1	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
5	2	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living things and appreciate biological diversity.	relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
5	3	Living Things and Environment	Nutrition in Animals	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe modes and processes of nutrition in animals at the grade-appropriate level. 2. Skills: The learner should be able to investigate, represent and explain ideas related to nutrition in animals using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate respect for living	The learner is guided to observe, explore and investigate nutrition in animals using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How are animal structures and behaviours adapted to obtaining food?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				things and appreciate biological diversity.					
5	4	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
5	5	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate,	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
6	1	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
6	2	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be	The learner is guided to observe, explore	How do plant structures and processes support reproduction and	KICD curriculum design; learner's book; teacher-prepared notes; charts/models;	Oral questions; observation; practical skills checklist; learner's notebook;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	survival?	locally available specimens; laboratory apparatus and safety equipment where appropriate.	diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
6	3	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	present explanations supported by evidence.				
6	4	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
6	5	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
7	1	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				plants and value the role of reproduction in plant survival.					
7	2	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
7	3	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate,	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
7	4	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
7	5	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be	The learner is guided to observe, explore	How do plant structures and processes support reproduction and	KICD curriculum design; learner's book; teacher-prepared notes; charts/models;	Oral questions; observation; practical skills checklist; learner's notebook;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	survival?	locally available specimens; laboratory apparatus and safety equipment where appropriate.	diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
8	1	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	present explanations supported by evidence.				
8	2	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
8	3	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
8	4	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				plants and value the role of reproduction in plant survival.					
8	5	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
9	1	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate,	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
9	2	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	The learner is guided to observe, explore and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do plant structures and processes support reproduction and survival?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
9	3	Living Things and Environment	Reproduction in Plants	At the end of the lesson, the learner should be	The learner is guided to observe, explore	How do plant structures and processes support reproduction and	KICD curriculum design; learner's book; teacher-prepared notes; charts/models;	Oral questions; observation; practical skills checklist; learner's notebook;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				able to: 1. Knowledge: The learner should be able to describe processes of sexual and asexual reproduction in plants. 2. Skills: The learner should be able to investigate, represent and explain ideas related to reproduction in plants using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should show care for plants and value the role of reproduction in plant survival.	and investigate reproduction in plants using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	survival?	locally available specimens; laboratory apparatus and safety equipment where appropriate.	diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
9	4	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	present explanations supported by evidence.				
9	5	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
10	1	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
10	2	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				responsibility for biodiversity and environmental conservation.					
10	3	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
10	4	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate,	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
10	5	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
11	1	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be	The learner is guided to observe, explore	How do organisms depend on one another and on their	KICD curriculum design; learner's book; teacher-prepared notes; charts/models;	Oral questions; observation; practical skills checklist; learner's notebook;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	environment?	locally available specimens; laboratory apparatus and safety equipment where appropriate.	diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
11	2	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	present explanations supported by evidence.				
11	3	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
11	4	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
11	5	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				responsibility for biodiversity and environmental conservation.					

Teacher's reflection on the term: _____

_____

TERM 3 — 40 planned curriculum lessons

Suggested school-calendar window: 9 weeks. Use remaining timetable periods for revision, CATs, practical work, projects, remediation, enrichment, examinations, feedback and school-based activities.

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
1	1	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
1	2	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge:	The learner is guided to observe, explore and investigate interdependence	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		equipment where appropriate.	exit ticket.	
1	3	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations,	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	supported by evidence.				
1	4	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
1	5	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and environmental conservation.	relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
2	1	Living Things and Environment	Interdependence of Life	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to explain relationships among organisms and their environment. 2. Skills: The learner should be able to investigate, represent and explain ideas related to interdependence of life using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate responsibility for biodiversity and	The learner is guided to observe, explore and investigate interdependence of life using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do organisms depend on one another and on their environment?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				environmental conservation.					
2	2	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
2	3	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
2	4	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
2	5	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge:	The learner is guided to observe, explore and investigate curved mirrors	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		apparatus and safety equipment where appropriate.	peer/self-assessment; exit ticket.	
3	1	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	supported by evidence.				
3	2	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
3	3	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
3	4	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				investigating light.					
3	5	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
4	1	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
4	2	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
4	3	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge:	The learner is guided to observe, explore and investigate curved mirrors	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		apparatus and safety equipment where appropriate.	peer/self-assessment; exit ticket.	
4	4	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	supported by evidence.				
4	5	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
5	1	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
5	2	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				investigating light.					
5	3	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
5	4	Force and Energy	Curved Mirrors	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to investigate image formation by curved mirrors and relate mirror properties to applications. 2. Skills: The learner should be able to investigate, represent and explain ideas	The learner is guided to observe, explore and investigate curved mirrors using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record	How does the shape of a mirror affect the image formed?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				related to curved mirrors using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should observe safety precautions and show curiosity when investigating light.	observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
5	5	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
6	1	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to	The learner is guided to observe, explore and investigate waves using suitable models, specimens,	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
6	2	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				safe participation during investigations.					
6	3	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
6	4	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	appropriate; discuss findings with peers; and present explanations supported by evidence.				
6	5	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
7	1	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials;	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
7	2	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
7	3	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
7	4	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations,	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	present explanations supported by evidence.				
7	5	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
8	1	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.				
8	2	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
8	3	Force and Energy	Waves	At the end of the lesson, the learner should be	The learner is guided to observe, explore	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models;	Oral questions; observation; practical skills checklist; learner's notebook;	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.		locally available specimens; laboratory apparatus and safety equipment where appropriate.	diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	
8	4	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

WK	LSN	STRAND	SUB-STRAND	LESSON LEARNING OUTCOMES	LEARNING EXPERIENCES	KEY INQUIRY QUESTIONS	LEARNING RESOURCES	ASSESSMENT METHODS	REFLECTION
				learner should demonstrate curiosity, cooperation and safe participation during investigations.	evidence.				
8	5	Force and Energy	Waves	At the end of the lesson, the learner should be able to: 1. Knowledge: The learner should be able to describe wave characteristics and investigate how waves transfer energy. 2. Skills: The learner should be able to investigate, represent and explain ideas related to waves using appropriate scientific procedures, observations, models and evidence. 3. Attitude: The learner should demonstrate curiosity, cooperation and safe participation during investigations.	The learner is guided to observe, explore and investigate waves using suitable models, specimens, diagrams or locally available materials; demonstrate the relevant process or procedure through a supervised activity; draw and label diagrams or record observations where appropriate; discuss findings with peers; and present explanations supported by evidence.	How do waves transfer energy from one place to another?	KICD curriculum design; learner's book; teacher-prepared notes; charts/models; locally available specimens; laboratory apparatus and safety equipment where appropriate.	Oral questions; observation; practical skills checklist; learner's notebook; diagrams; written exercises; group tasks; peer/self-assessment; exit ticket.	

Teacher's reflection on the term: _____

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Practical work, safety and assessment guidance

Plan practical activities before the lesson. Check apparatus, materials and classroom/laboratory safety. Use teacher demonstrations where learner handling would be unsafe or facilities are limited. Do not taste laboratory materials. Wear appropriate protective equipment, maintain clear work areas and dispose of materials responsibly.

Use continuous assessment evidence: oral questioning, observation, labelled diagrams, learner records, practical performance, group investigations, projects, written exercises and end-of-topic tests. Provide feedback and targeted remediation or enrichment.

Reference

Kenya Institute of Curriculum Development (KICD), Grade 9 Integrated Science Curriculum Design (2024). Official curriculum design: https://kicd.ac.ke/wp-content/uploads/2024/06/Integrated-Science-Grade-9-Design-Formatted-April-2024.pdf

Teacher-prepared original scheme. Strand/sub-strand lesson allocations follow the KICD Grade 9 design summary; lesson outcomes, learning experiences, inquiry questions and resource suggestions are original planning formulations for adaptation.